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数据库实验报告 第11周

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实验内容

由于每次实验前我都有恢复数据库的习惯，这次恢复的太急，没看见要求就恢复了，将原先的Worker表删除了，现在新建Worker表：

```
Use School
create table Worker (
    Number char(5),
    Name char(8) constraint U1 unique,
    Sex char(1),
    Sage int constraint U2 check (Sage <= 28),
    Department char(20),
    constraint PK_Worker primary key (Number)
)
insert into Worker values('0001', '李强', 'M', 26, 'CS')
select * from Worker
```

%					
结果	消息				
	Number	Name	Sex	Sage	Department
	0001	李强	M	26	CS

1.创建触发器T4如下，因为Worker表中包含inserted表中的数据，因此要将其排除：

```
SQLQuery1.sql - (...CAK\86135 (52))*  X
[Use School
[Use School
go
[create trigger T4 on Worker
for insert
as
[if (select sage from inserted) <= (select max(sage) from Worker where Number not in (select Number from inserted))
[begin
print 'T4'
Rollback transaction
end

100 %
消息
命令已成功完成。
```

2.尝试插入如下记录，其sage小于表中的最大值（26），插入失败。

```
SQLQuery1.sql - (...CAK\86135 (52))*  X
[Use School
insert into Worker values('0002', '张强', 'M', 24, 'CS')

100 %
消息
T4
消息 3609, 级别 16, 状态 1, 第 2 行
事务在触发器中结束。批处理已中止。
```

3.创建触发器如下：

```
SQLQuery1.sql - (...CAK\86135 (52))*
Use School
go
create trigger T5 on Worker
for update
as
if update(sage) and EXISTS (select 1 from inserted i join deleted d on i.Number = d.Number where i.sage <= d.sage)
begin
print 'hhhh'
Rollback transaction
end
```

100 %

消息

命令已成功完成。

完成时间: 2024-11-23T15:24:02.0783459+08:00

4.尝试更新记录, 将sage设置为1 (原先sage为26) , 失败:

```
SQLQuery1.sql - (...CAK\86135 (52))*
update Worker set Sage = 1 where Number = '0001'
```

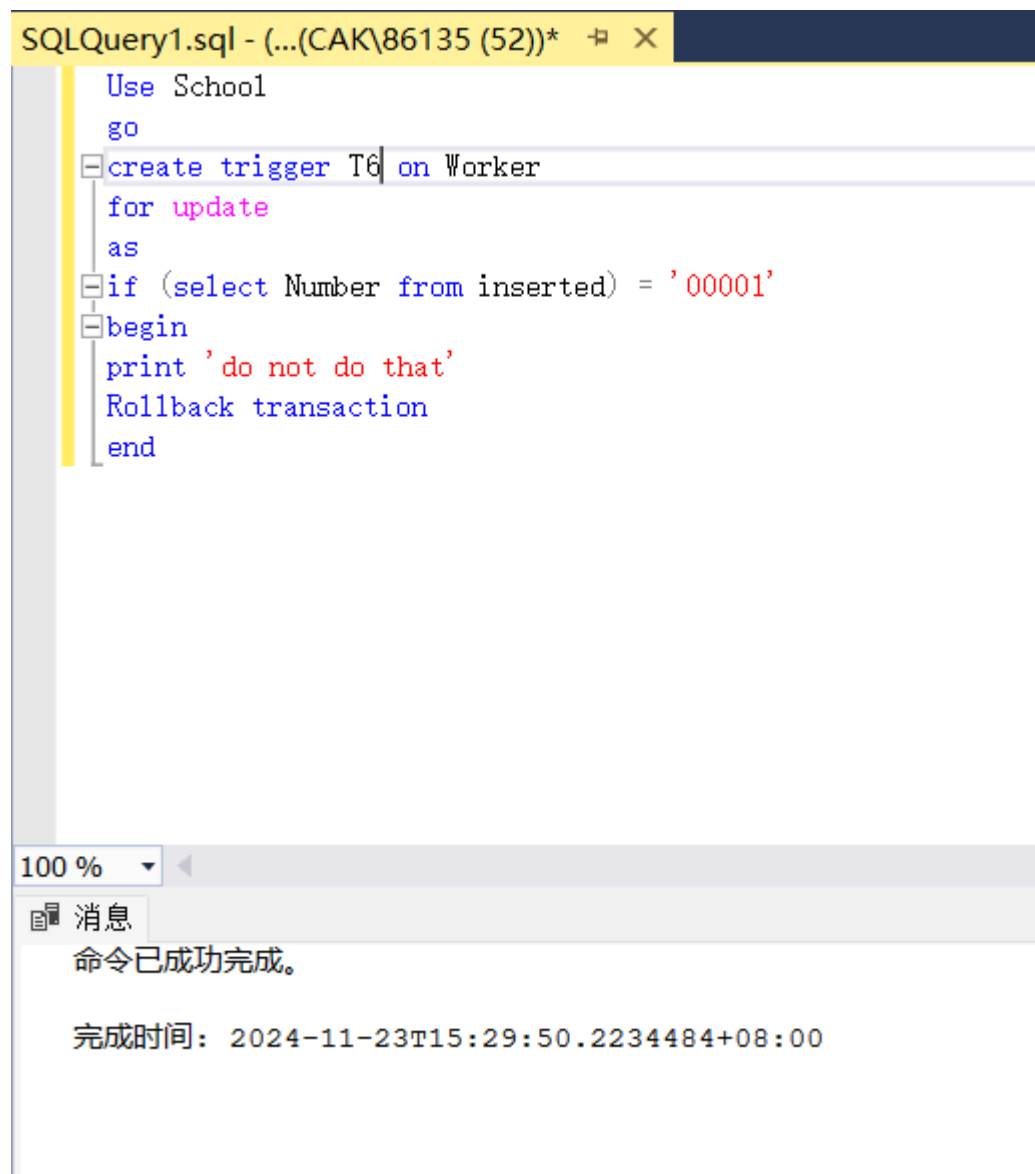
100 %

消息

hhhh
消息 3609, 级别 16, 状态 1, 第 2 行
事务在触发器中结束。批处理已中止。

完成时间: 2024-11-23T15:25:57.8844163+08:00

6.创建触发器如下:



The screenshot displays the SQL Server Enterprise Manager interface. At the top, a tab is labeled "SQLQuery1.sql - (...CAK\86135 (52))*". The main window contains a SQL script with the following text:

```
Use School
go
create trigger T6 on Worker
for update
as
if (select Number from inserted) = '00001'
begin
print 'do not do that'
Rollback transaction
end
```

Below the script editor, there is a zoom level dropdown set to "100 %". At the bottom, a "消息" (Messages) pane shows a status message:

命令已成功完成。

完成时间: 2024-11-23T15:29:50.2234484+08:00

首先插入一个Number = '00001' 的记录:

☐ Use School

```
insert into Worker values('00001', '张强', 'M', 27, 'CS')  
select * from Worker
```

100 %

结果

消息

	Number	Name	Sex	Sage	Department
1	00001	张强	M	27	CS
2	0001	李强	M	26	CS

然后尝试对其修改，违反触发器T6，失败：

SQLQuery1.sql - (...CAK\86135 (52))*

Use School
update Worker set sex = 'F' where Number = '00001'

100 %

消息

do not do that
消息 3609, 级别 16, 状态 1, 第 2 行
事务在触发器中结束。批处理已中止。

完成时间: 2024-11-23T16:02:35.4027764+08:00

6.首先创建Stu_Card表:

```
Use School
create table Stu_Card(
  card_id char(14),
  stu_id char(10),
  remained_money decimal (10,2),
  constraint PK_stu_card primary key(card_id),
  constraint FK_stu_card foreign key (stu_id) references STUDENTS (sid) on delete cascade
)
insert into Stu_Card values('05212567','800001216',100.25)
insert into Stu_Card values('05212222','800005753',200.50)
select * from Stu_Card
```

100 %

结果 消息

	card_id	stu_id	remained_money
1	05212222	800005753	200.50
2	05212567	800001216	100.25

创建视图，该视图关联多个表，是不可更新的。

```
create view S_ as
select s.sid, s.sname, c.card_id
from STUDENTS s ,Stu_Card c
where s.sid = c.stu_id
```

100 %

消息

命令已成功完成。

完成时间：2024-11-23T16:37:01.8642543+08:00

SQLQuery1.sql - (... (CAK\86135 (52))*

```
select * from S_
```

100 %

结果 消息

	sid	sname	card_id
1	800005753	waqcj	05212222
2	800001216	gfxrgs	05212567

SQLQuery1.sql - (... (CAK\86135 (52))*

```
insert into S_ values ('1','1','1')
```

100 %

消息

消息 4405, 级别 16, 状态 1, 第 2 行
视图或函数 's_' 不可更新, 因为修改会影响多个基表。

完成时间: 2024-11-23T16:38:24.8476471+08:00

创建触发器如下, 使用INSTEAD OF 触发器重新定义insert操作:

SQLQuery1.sql - (...CAK\86135 (52))*

```
Use School
go
create trigger T7 on S_
instead of insert
as
begin
    set nocount on
    if not exists (
        select s.sid
        from students s join inserted i on i.sid = s.sid
    )
    begin
        insert into STUDENTS
            select sid, sname, null, null from inserted
        insert into Stu_Card
            select card_id, sid, null from inserted
    end
    else print '数据已经存在'
end
```

100 %

消息

命令已成功完成。

完成时间: 2024-11-23T16:40:26.9138724+08:00

再次执行插入操作，插入成功：

SQLQuery1.sql - (...CAK\86135 (52))*

```
insert into S_ values ('1', '1', '1')
```

100 %

消息

(1 行受影响)

完成时间: 2024-11-23T16:40:56.2103153+08:00

查看表，发现各表已经按重新定义的insert操作成功插入数据：

```

select * from STUDENTS where sid = '1'
select * from Stu_Card where stu_id = '1'
select * from S

```

100 %

 结果
  消息

	sid	sname	email	grade
1	1	1	NULL	NULL

	card_id	stu_id	remained_money
1	1	1	NULL

	sid	sname	card_id
1	800005753	waqcj	05212222
2	800001216	gfxrgs	05212567
3	1	1	1